Example 8.9

Ammu has five toys which are identical and one of them is underweight. Her sister, Harini, chooses one of these toys at random. Find the probability for Harini to choose an underweight toy?

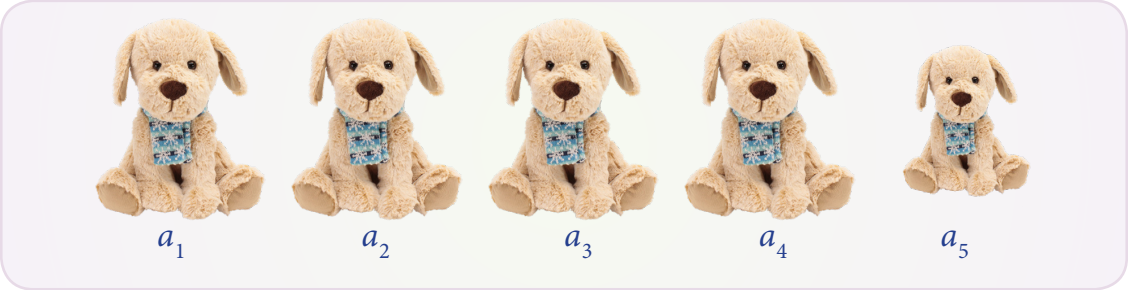


fig. 8.7 Identical toys with on underweight toy

Solution :

It is seen from fig. 8.7, the sample space is S = $\{a_1, a_2, a_3, a_4, a_5\}$.Define the events A_1, A_2, A_3, A_4 and A_5 as

A : Harini chooses the underweight toy

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{5}$$

Therefore, the probability for Harini to choose an underweight toy is 1/5.

Example 8.10

A box contains 3 red and 4 blue socks. Find the probability of choosing two socks of same colour.



= 7 socks (3 Red + 4 Blue)

i.e., $n = 7$

A_1 : choosing 2 socks from 3 socks

$n(A_1) = 3C_2$ ways = 3 ways

A_2 : choosing 2 socks from 4 socks

$n(A_2) = 4C_2$ ways = 6 ways

fig.8.8 choosing 2 socks of same colour

Solution :

From fig. 8.8, total number of socks = 3 + 4 = 7

If two socks are drawn at random, then

No. of ways of selecting 2 socks = $7C_2 = 21$